

Chapter 117

Shape Nodes

This chapter details the Shape nodes available in Fusion.

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sBoolean



The sBoolean node

The sBoolean node combines or excludes overlapping areas of two shapes based on a menu of boolean operations.

Like almost all shape nodes, you can only view the sBoolean node's results through a sRender node.

External Inputs

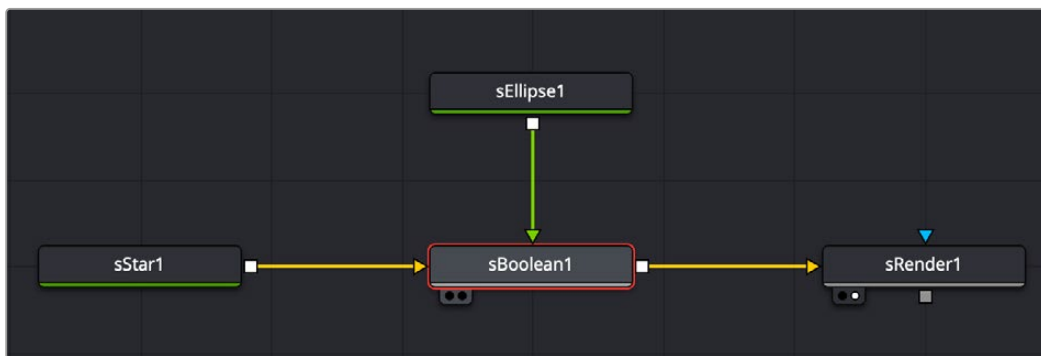
The following inputs appear on the node's tile in the Node Editor. Except when using the subtract boolean operation, which shape you connect into which input does not matter.

Input1: [orange, required] This input accepts the output of another shape node. This input is used as the base shape when the subtract boolean operation is chosen.

Input2: [green, optional] This input accepts the output of another shape node. This input is used to cut the base shape hole when the subtract boolean operation is chosen.

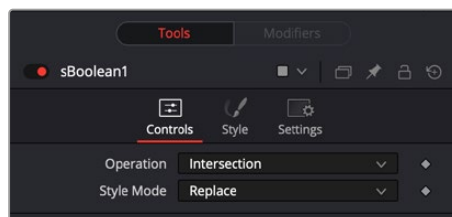
Basic Node Setup

The sBoolean node is used to combine two shape nodes. The output of the sBoolean can then be output to another shape node or a sRender node for viewing.



Star and ellipse shapes combined in an sBoolean node, then output to an sRender for viewing and further processing

Inspector



The sBoolean Controls tab

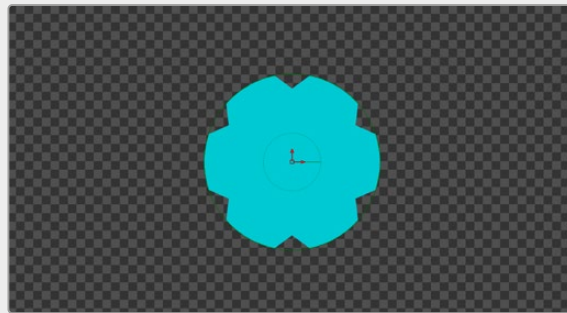
Controls

The Controls tab is primarily used to select the boolean operation that determines how the two shapes are combined.

Operation

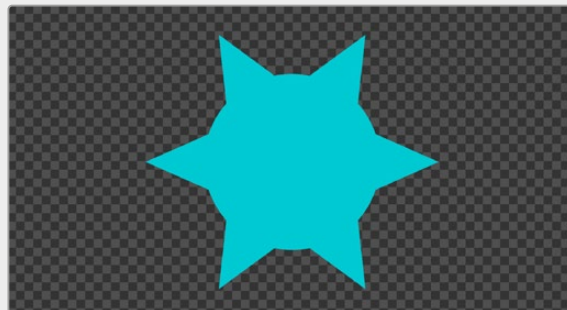
The operation menu includes four boolean operations:

Intersection: Sometimes called an AND operation, this setting will only show areas where the two shapes overlap. The result is only where input 1 AND input 2 overlap.



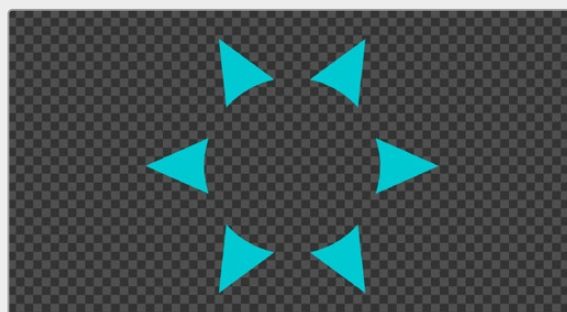
Star and ellipse shapes with an sBoolean node set to intersection

Union: Sometimes called an OR operation, this setting will only show areas where either of the two shapes exists. The result is where either input 1 OR input 2 exists. The Union setting is similar to the result of the sMerge node.



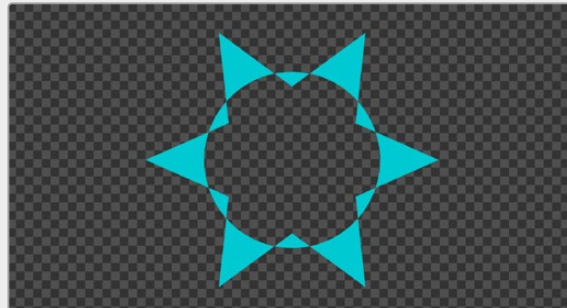
Star and ellipse shapes with an sBoolean node set to union

Subtract: Sometimes called a NOT operation, this setting outputs the shape of input 1 but eliminates the areas where input 2 overlaps. The result is input 1 minus input 2.



Star and ellipse shapes with an sBoolean node set to subtract

Xor: Sometimes called an AND NOT operation, this setting outputs the shape of input 1 or input 2 but eliminates the areas where they overlap. The result is (input 1 minus input 2) + (input 2 minus input 1).

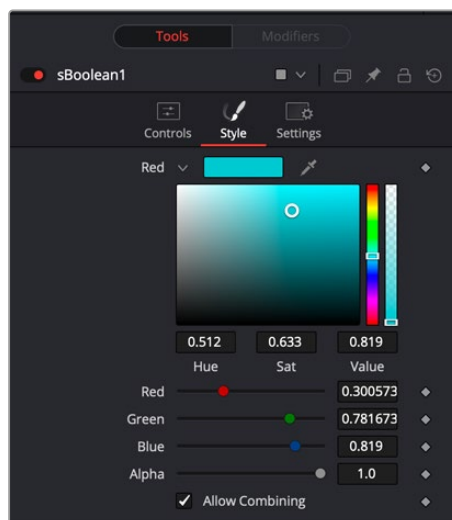


Star and ellipse shapes with an sBoolean node set to xor

Style Mode

The Style mode menu only includes one option. The Replace setting replaces the color and alpha level of the incoming shapes with the color set in the Style tab.

Style Tab



The sBoolean Style tab

Style

Any color assigned to the individual shape nodes is replaced by the color set using the Style tab controls.

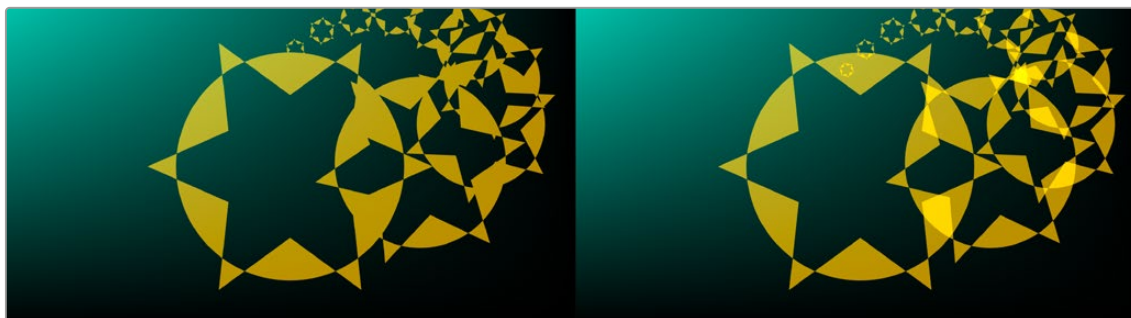
Color

The color controls determine the color of the output shape from the sBoolean node. To choose a shape color, you can click the color disclosure arrow, use the color swatch, or drag the eyedropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter each color channel's value or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself.

For instance, if an ellipse's Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a duplicate or grid node that causes the shape to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area. When using the sBoolean node, the individual shape node checkboxes are ignored, and the sBoolean node's checkbox determines the Alpha channel's behavior.



Allow Combining Enabled (left), Allow Combining Disabled (right)

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sBSpline [sBSp]

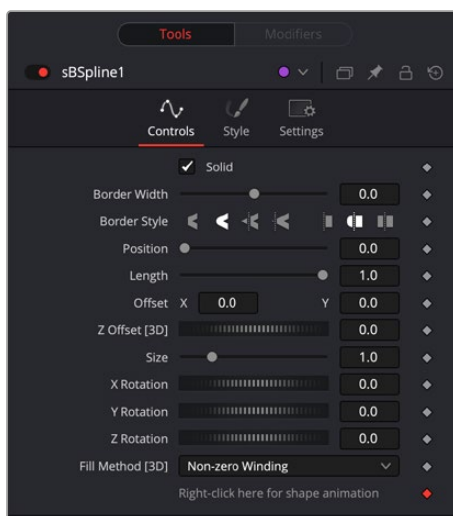


The sBSpline node

Introduction

sBSpline is identical to a sPolygon in all respects except one. Where sPolygon uses Bézier splines, this node uses B-Splines. Where Bézier splines employ a central point and two handles to manage the smoothing of the spline segment, a B-Spline requires only a single point. This means that a sBSpline shape requires far fewer control points to create a nicely smoothed shape.

Usage



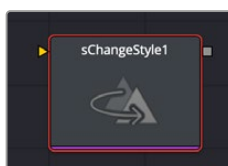
The sBSpline controls

When first added to a Node Editor, the sBSpline shape consists of only Center control, which is visible onscreen.

Points are added to the sBSpline by clicking in the viewer. Each new point is connected to the last one created, but instead of the spline going directly through each control point, sBSpline control points only influence the spline shape. The control point pulls the spline in its direction to create a smooth curve.

Like the sPolygon shape tool, the sBSpline shape auto-animates. Adding this node to the Node Editor adds a keyframe to the current frame. Moving to a new frame and changing the shape creates a new keyframe and interpolate between the two defined shapes.

sChangeStyle [sCs]



The sChangeStyle node

sChangeStyle Node Introduction

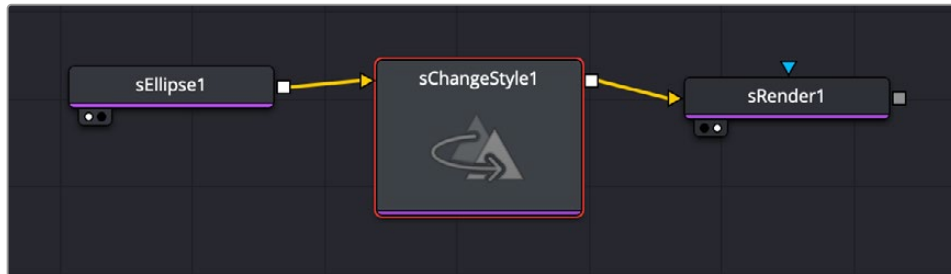
The sChangeStyle node assigns a new style to the combined shape tree, overriding the style (color and Allow Combining settings) set in the upstream shape tools.

Inputs

The single input on the sChangeStyle tool is used to connect the Shape node you want to change the style of.

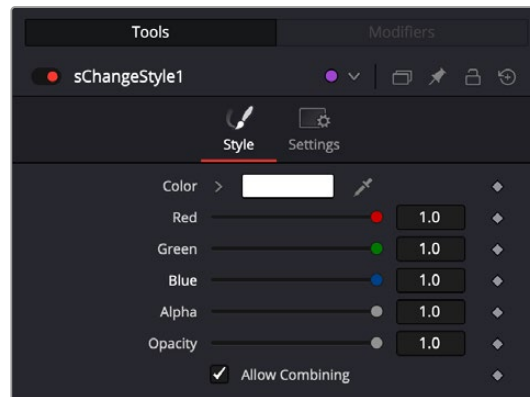
Basic Node Setup

Connect the shape output that you wish change the style of to the input. The output of the node then gets passed to the remainder of the shape node tree.



A simple node tree showing the sChangeShape node taking the output of an sEllipse shape, performing the style change, and passing it to the sRender node.

Inspector



The sChangeStyle controls

Controls

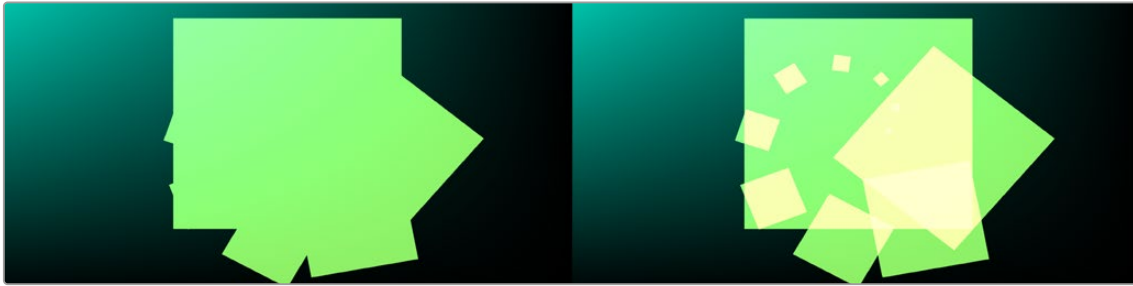
The Style tab is used to assign color to the shape and control its transparency.

Color

The Color parameter controls determine the color of the fill and border from the sPolygon node. To choose a shape color, you can click the color disclosure arrow and use the color swatch, or drag the eye dropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter the value of each color channel or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained, even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself. For instance, if a rectangle Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value, even if the shape passes through a duplicate or grid node that causes the shape and Alpha channel to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.



Allow Combining enabled (left), Allow Combining disabled (right)

sDuplicate



The sDuplicate node

The sDuplicate node creates copies of the input shape, offsetting each copy's position, size, and rotation. Like almost all shape nodes, you can only view the sDuplicate node's results through a sRender node.

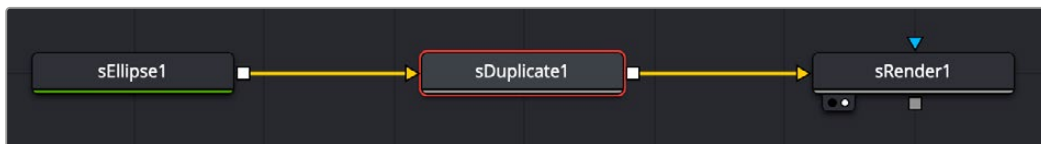
External Inputs

The following input appears on the node's tile in the Node Editor:

Input1: [orange, required] This input accepts the output of another Shape node. The shape connected to this input is copied and offset based on the controls in the Inspector.

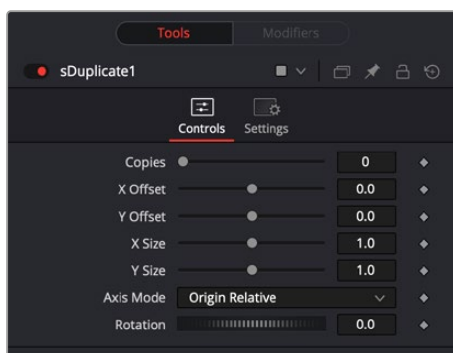
Basic Node Setup

The sDuplicate node takes a single input that is most often from a single or compound shape from a sMerge or sBoolean node. The sDuplicate node creates copies of the incoming shape and offsets them to create a pattern. The output of the sDuplicate can then be output to another shape node or to a sRender node for viewing or compositing into the greater node tree.



An sEllipse shape connected to an sDuplicate and then output to an sRender for viewing and combining with other elements

Inspector



The sDuplicate Controls tab

Controls

The Controls tab is used to determine the number of copies and set their position, size, and rotation offset.

Copies

This slider determines the number of copies created by the node. The number does not include the original shape, so entering a value of five will produce five copies plus the original.

X and Y Offset

These sliders set the X and Y distance between each of the copies. Each copy is offset from the previous copy by the value entered in the X and Y number fields. The copies all start at 0, the center of the original shape, and are offset from there. Using Fusion's normalized coordinate system, entering X Offset at 0.5 would move each copy half the frame's width to the right. Entering -1.0 would move each copy to the left by the width of the frame.

X and Y Size

Sets the X and Y size offset based on the previous shape size. For instance, an X and Y value of 1.0 creates copies identical in size to the original but. Entering a value of X and Y of 0.5 will cause each copy to be half the size of the copy before it.

Axis Mode

The Axis mode menu provides four options for determining how each copy determines its rotational pivot point.

- **Absolute:** Allows you to set an X and Y position for the axis of rotation based on the original shape's location. The axis of rotation is then copied and offset with each duplicated shape.
- **Origin Relative:** Each copy uses its center point as its axis of rotation.
- **Origin Absolute:** Each copy uses the center of the original shape as its axis of rotation.
- **Progressive:** Compounds each shape copy by progressively transforming each copy based on the previous shape's position, rotation, and scale.

X and Y Pivot

The X and Y pivot controls are displayed when the Axis mode is set to Absolute. You can use these position controls to place the axis of rotation.

Rotation

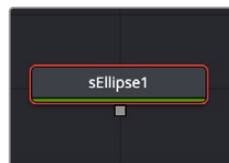
Determines an offset rotation applied to each copy. The rotation is calculated from the offset rotation of the previous copy. To rotate all copies identically, use the Angle parameter on the original shape or use a sTransform node.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sEllipse



The sEllipse node

The sEllipse node is used to create circular shapes. Like almost all shape nodes, you can only view the sEllipse node's results through a sRender node.

External Inputs

This node generates shapes and does not have any inputs.

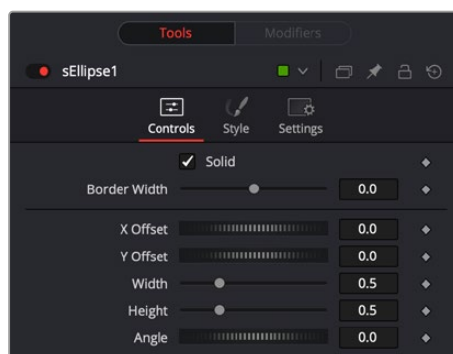
Basic Node Setup

The sEllipse node is a shape generator, meaning it generates a shape and therefore has no input. The output of the sEllipse can go into a sRender for viewing and further compositing or, more likely, connect to another shape node like sGrid or sDuplicate.



An sEllipse node connecting to an sGrid node, and then viewed using an sRender node

Inspector



The sEllipse Controls tab

Controls

The Controls tab is used to define the elliptical shape characteristics, including fill, border, size, and position.

Solid

When enabled, the Solid checkbox fills the elliptical shape with the color defined in the Style tab. When disabled, an outline created by the Border Width control is displayed, and the center is made transparent.

Border Width

This parameter expands or contracts the border around the shape. Although it can be used when the Solid checkbox is enabled, it is primarily used to determine the outline thickness when the checkbox is disabled.

Cap style

When the Solid checkbox is disabled, three cap style options are displayed. The cap styles can create lines with flat, rounded, or squared ends. Flat caps have flat, squared ends, while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The position parameter is only displayed when the Solid checkbox is disabled. It allows you to position the starting point of the shape. When used in conjunction with the length parameter, it positions the gap in the ellipse outline.

Length

The length parameter is only displayed when the Solid checkbox is disabled. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the length parameters allows you to create write-on style animations.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. An X offset of 0.0 is centered, and a value of 0.5 places the center of the shape directly on the right edge of the frame.

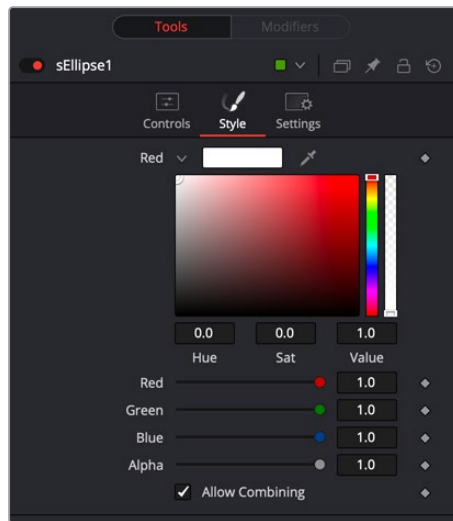
Width/Height

The width and height determine the vertical and horizontal size of the ellipse. If the values are identical, then you have a perfect circle.

Angle

The angle rotates the shape, which on a perfect circle doesn't change the image all that much, but if you create an oval or an outline with a short length, you can rotate the shape based on the center axis.

Style Tab



The sEllipse Style tab

Style

The Style tab is used to assign a color to the shape and control its transparency.

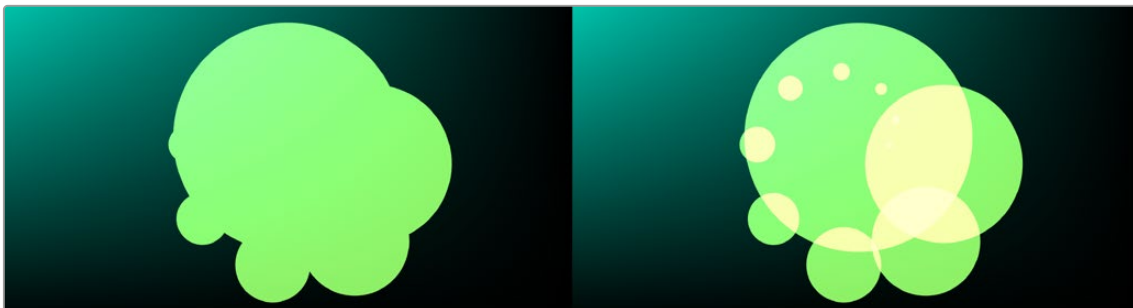
Color

The color controls determine the color of the fill and border. To choose a shape color, you can click the color disclosure arrow, use the color swatch, or drag the eyedropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter each color channel's value or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself.

For instance, if an ellipse's Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a Duplicate or Grid node that causes the shape to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.



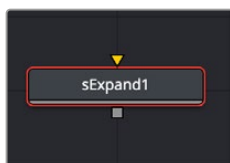
Allow Combining Enabled (left), Allow Combining Disabled (right)

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sExpand



The sExpand node

The sExpand node is used to dilate or erode shapes. Like almost all Shape nodes, you can only view the sExpand node's results through a sRender node.

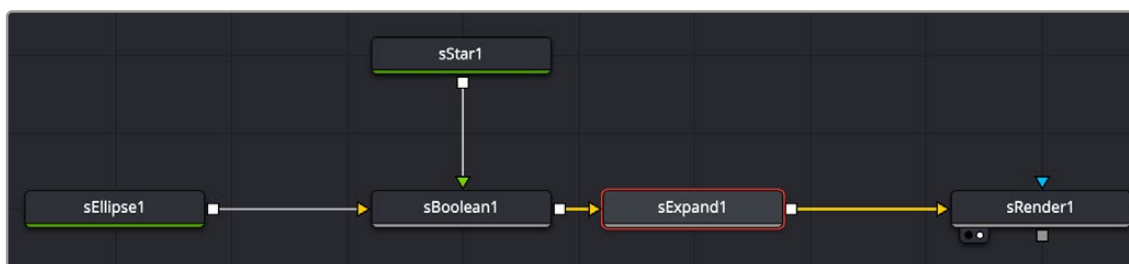
External Inputs

The following input appears on the node's tile in the Node Editor.

Input1: [orange, required] This input accepts the output of another shape node. This shape or compound shape connected to this input is either eroded or dilated.

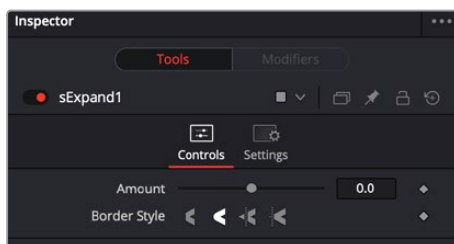
Basic Node Setup

The sExpand node takes a single input that is most often from a compound shape. However, it can be used on single shapes like sStars and sNgons. The output of the sExpand can then be output to another shape node or to a sRender node for viewing or compositing into the greater node tree.



Star and ellipse shapes combined in an sBoolean node, then output to an sExpand for dilating or eroding

Inspector



The sExpand Controls tab

Controls

The Controls tab includes all of the parameters for the sExpand node.

Amount

A positive value dilates the shape while a negative value erodes it.

Border Style

The border style controls how the expanded or contracted shapes join at the corners. There are four styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter and Miter Clip maintain pointed edges, until a certain threshold. The Threshold is set by the Miter limit slider.

Miter Limit

The Miter parameter is only displayed when the Miter or Miter Clip border style is selected. The miter limit determines when the pointed edges become beveled based on the shape's thickness.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sGrid



The sGrid node

The sGrid node replicates the shape on an X and Y grid and adds the ability to offset the rows and columns. Like almost all Shape nodes, you can only view the sGrid node's results through a sRender node.

External Inputs

The following input appears on the node's tile in the Node Editor.

Input1: [orange, required] This input accepts the output of another Shape node. The shape connected to this input is replicated on a custom grid.

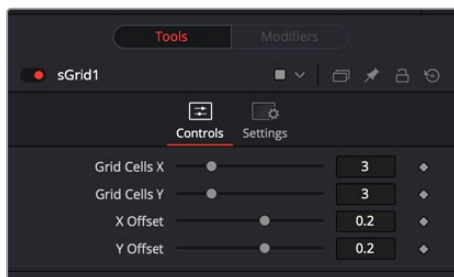
Basic Node Setup

The sGrid node takes a single input from a single or compound shape from an sMerge or sBoolean node. The sGrid node places the incoming shape on a grid of rows and columns. The output of the sGrid can then be output to another Shape node or a sRender node for viewing or compositing into the greater node tree.



An sEllipse shape connected to an sGrid and then output to an sRender for viewing and combining with other elements

Inspector



The sGrid Controls tab

Controls

The Controls tab is used to determine the number of grid cells and their offset position.

Grid Cells X and Y

These parameters set the number of cells on the grid, both horizontally and vertically. For instance, entering 5 in the X and Y number field creates five rows of the shape and five columns.

X and Y Offset

Sets the X and Y distance between the rows and columns. An offset value of 0.0 will have all the rows and columns on top of each other. Entering X Offset at 1.0 would spread the columns the width to the frame.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sJitter



The sJitter node

The sJitter node is most often used to randomly position an array of shapes generated from a sGrid or sDuplicate node. However, it includes an auto-animating random mode that can be used to distort and randomly jitter single shapes.

Like almost all Shape nodes, you can only view the sJitter node's results through a sRender node.

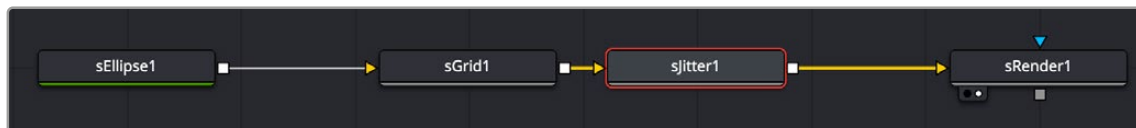
External Inputs

The following input appears on the node's tile in the Node Editor.

Input1: [orange, required] This input accepts the output of another Shape node. The shape connected to this input is offset, distorted, and animated based on the sJitter node settings.

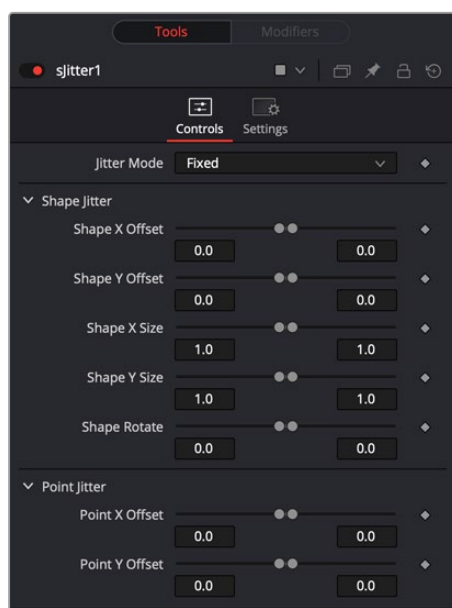
Basic Node Setup

The sJitter node takes an array of shapes from a sGrid or sDuplicate node and randomly changes their position, size, and rotation. The output of the sExpand can then be output to another Shape node or to a sRender node for viewing or compositing into the greater node tree.



An array of shapes created by the sGrid node input into an sJitter node to randomly offset or scale the shapes

Inspector



The sJitter Controls tab

Controls

The Controls tab offers range sliders that determine the variation amount for offset, size, and rotation. The Point Jitter parameters are used to offset the invisible points that create the vector shapes.

Jitter Mode

The Jitter Mode menu allows you to choose between static position and size offsets or enabling an auto-animation mode. Leaving the default Fixed selection allows you to offset a grid of shapes, animating with keyframes or modifiers if needed. The Random menu selection auto-animates the parameters based on the range you define using the range sliders. If all the range sliders are left in the default position, no random animation is created. Increasing the range on any given parameter will randomly animate that parameter between the range slider values.

Shape X and Y Offset

These parameters set the horizontal and vertical offset from the shape array's original position. This is done randomly, so not all shapes in the array will offset by the same amount.

Shape X and Y Size

These parameters set the horizontal and vertical scaling for each shape in an array. The left range value decreases the scale, and the right range value increases the scale. This is done randomly, so not all shapes in the array will scale by the same amount.

Shape Rotate

This parameter rotates each shape in an array.

Point Jitter

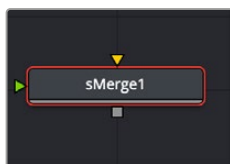
The X and Y Point Jitter parameters use the vector control points to distort the shape. This can be used to give a distressed appearance to ellipses or wobbly animation to other shapes.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

sMerge



The sMerge node

The sMerge node combines shapes similar to a standard Merge node, except the sMerge node can accept more than two shape inputs.

Like almost all Shape nodes, you can only view the sMerge node's results through a sRender node.

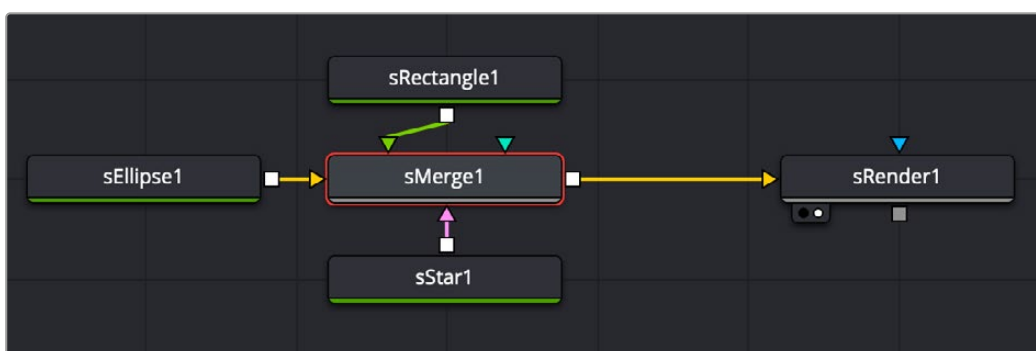
External Inputs

The node displays only two inputs first, but as each shape node is connected, a new input appears on the node, assuring there is always one free to add a new shape into the composite.

Input[#]: These multi-colored inputs are used to connect multiple Shape node. There is no limit to the number of inputs this node can accept. The node dynamically adds more inputs as needed, ensuring that there is always at least one input available.

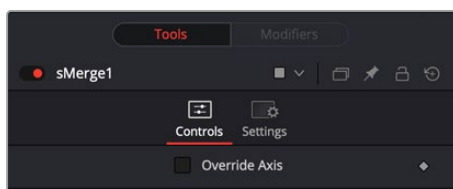
Basic Node Setup

The sMerge node is used to combine two Shape nodes. In terms of layering, each subsequent input is placed over the previous input. For instance, the first shape connected to the orange input is the bottom-most shape, the green input is layered over it, and if a third shape is connected to the pink input, that is the topmost layer.



Three shapes are combined in an sMerge node, then output to an sRender for viewing and further processing.

Inspector



The sMerge Controls tab

Controls

The only control for the sMerge node is the Override Axis checkbox, which overrides the shape's axis.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

sNGon



The sNGon node

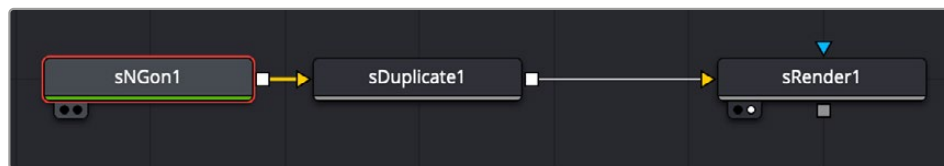
The sNGon node is used to create multi-sided shapes like triangles, pentagons, and octagons. Like almost all Shape nodes, you can only view the sNGon node's results through a sRender node.

External Inputs

This node generates shapes and does not have any inputs.

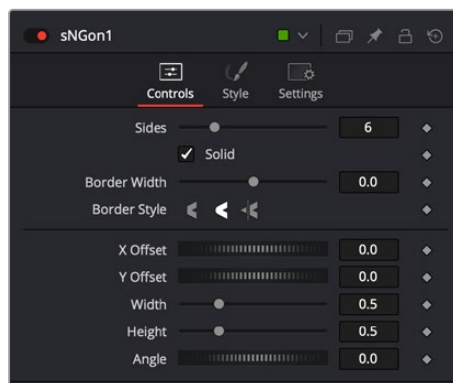
Basic Node Setup

The sNGon node is a shape generator, meaning it generates a shape and therefore has no inputs. The output of the sNGon can go into a sRender for viewing and further compositing or, more likely, connect to another Shape node like sGrid or sDuplicate.



An sNGon node connecting to an sDuplicate node, and then viewed using an sRender node.

Inspector



The sNGon Controls tab

Controls

The Controls tab is used to define multi-sided shape characteristics, including fill, border, size, and position.

Solid

When enabled, the Solid checkbox fills the NGon shape with the color defined in the Style tab. When disabled, an outline created by the Border Width control is displayed, and the center is made transparent.

Border Width

This parameter expands or contracts the border around the shape. Although it can be used when the Solid checkbox is enabled, it is primarily used to determine the outline thickness when the checkbox is disabled.

Border Style

The Border Style parameter controls how the sides of the NGon join at the corners. There are three styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter maintains pointed corners.

Cap style

When the Solid checkbox is disabled, three cap style options are displayed. The cap styles can create lines with flat, rounded, or squared ends. Flat caps have flat, squared ends, while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The Position parameter is only displayed when the Solid checkbox is disabled. It allows you to position the starting point of the shape. When used in conjunction with the Length parameter, it positions the gap in the outline.

Length

The Length parameter is only displayed when the Solid checkbox is disabled. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the Length parameters allows you to create write-on style animations.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. So, an X offset of 0.0 is centered and a value of 0.5 places the center of the shape directly on the right edge of the frame.

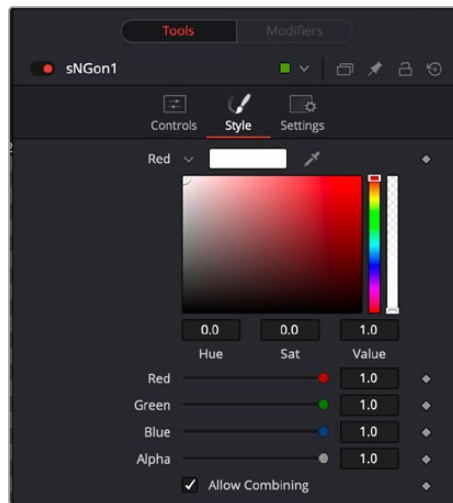
Width/Height

The Width and Height parameters determine the vertical and horizontal size of the ellipse. If the values are identical, then all sides are of equal length.

Angle

The Angle parameter rotates the shape based on the center axis.

Style Tab



The sNGon Style tab

Style

The Style tab is used to assign a color to the shape and control its transparency.

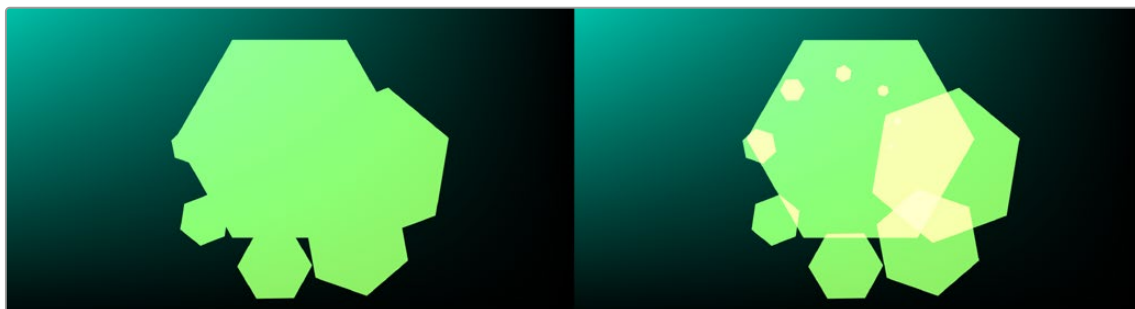
Color

The Color controls determine the color of the fill and border. To choose a shape color, you can click the color disclosure arrow, use the color swatch, or drag the eyedropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter each color channel's value or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself.

For instance, if a NGon Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a duplicate or grid node that causes the shape to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.

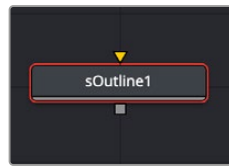


Allow Combining Enabled (left), Allow Combining Disabled (right)

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sOutline



The sOutline node

The sOutline node is used to create outlines from merged or boolean compound shapes. The individual shapes retain their own style, color, size, position, and other characteristics. The only difference is the border thickness, border style, position, and length are applied to all incoming shapes uniformly in the sOutline node.

Like almost all shape nodes, you can only view the sOutline node's results through a sRender node.

External Inputs

The following input appears on the node's tile in the Node Editor:

Input1: [orange, required] This input accepts the another shape node's output, but more likely a compound shape from a sMerge or sBoolean. An outline is created from the compound shape connected to this input.

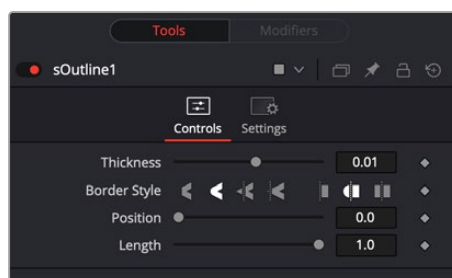
Basic Node Setup

The sOutline node takes a single input that is most often from a compound shape, however, it can sometimes be useful on single shapes to create double outlines. The output of the sOutline can then be output to another Shape node or to a sRender node for viewing or compositing into the greater node tree.



A compound shape from an sBoolean node is connected to an sOutline for creating a complex outlined shape

Inspector



The sOutline Controls tab

Controls

The Controls tab is used to define the outline thickness, border and cap style, position, and length that is applied to the compound shape connected to the input.

Thickness

This parameter controls the width of the outline.

Border Style

The Border Style parameter controls how the outline joins at the corners. There are three styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter maintains pointed corners.

Cap style

Three Cap Style options are used to create lines with flat, rounded, or squared ends. Flat caps have flat, squared ends, while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The Position parameter allows you to position the starting point of the shape. When used in conjunction with the Length parameter, it positions the gap in the outline.

Length

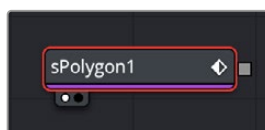
The Length parameter controls the end position of the outline. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the Length parameters allows you to create write-on style animations.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sPolygon [sPly]



The sPolygon node

The sPolygon tool allows users to draw custom shapes. Through intuitive point manipulation, this tool allows users to effortlessly generate intricate forms, curves, and lines, which tie in with the existing shape tools in Fusion. It provides the freedom to design custom elements, making it a valuable tool in motion graphics design.

Inputs

This node generates shapes and does not have any inputs.

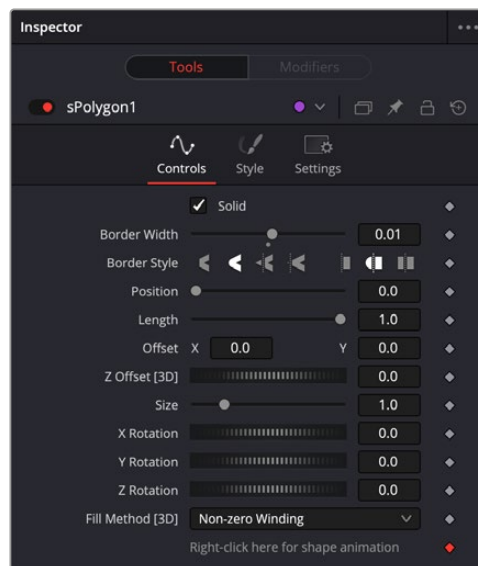
Basic Node Setup

The sPolygon node is a shape generator, meaning it generates a shape and therefore has no inputs. The output of the sPolygon can go into a sRender for viewing and further compositing or, more likely, connect to another Shape node like sGrid or sDuplicate.



An sPolygon node connecting to an sDuplicate node, and then viewed using an sRender node

Inspector



The sPolygon Controls tab

Controls

The Controls tab is used to define the polygon characteristics, including fill, border, size, and position.

Solid

When enabled, the Solid checkbox fills the rectangle shape with the color defined in the Style tab. When disabled, an outline created by the Border Width control is displayed, and the center is made transparent.

Border Width

This parameter expands or contracts the border around the shape. Although it can be used when the Solid checkbox is enabled, it is primarily used to determine the outline thickness when the checkbox is disabled.

Border Style

The Border Style parameter controls how the sides of the rectangle join at the corners. There are three styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter maintains pointed corners.

Cap style

The cap styles can create lines with flat, rounded, or squared ends. Flat caps have flat, squared ends, while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The Position parameter is only displayed when the Solid checkbox is disabled. It allows you to position the starting point of the shape. When used in conjunction with the Length parameter, it positions the gap in the outline.

Length

The Length parameter is only displayed when the Solid checkbox is disabled. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the Length parameters allows you to create write-on style animations.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. So an X offset of 0.0 is centered and a value of 0.5 places the center of the shape directly on the right edge of the frame.

Z Offset [3D]

If used with the 3D toolset, you can offset the polygon in back and forth along the Z-axis by adjusting these controls.

Size

Use the Size control to adjust the scale of the polygon shape, without affecting the relative behavior of the points that compose the shape or setting a keyframe in the shape animation.

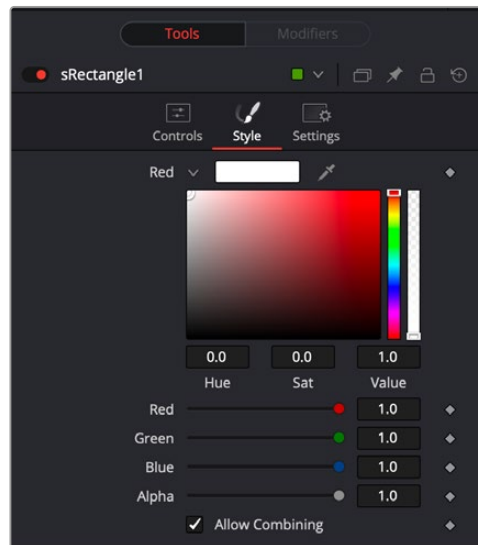
X,Y,Z Rotation

Use these three controls to adjust the rotation angle of the shape along any axis.

Fill Method [3D]

The Fill Method menu offers two different techniques for dealing with overlapping regions of a polyline. If overlapping segments in a shape are causing undesirable holes to appear, try switching the setting of this control from Alternate to Non Zero Winding.

Style Tab



The sPolygon Style tab

Style

The Style tab is used to assign color to the shape and control its transparency.

Color

The Color parameter controls determine the color of the fill and border from the sPolygon node. To choose a shape color, you can click the color disclosure arrow and use the color swatch, or drag the eye dropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter the value of each color channel or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained, even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself. For instance, if a rectangle Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a duplicate or grid node that causes the shape and Alpha channel to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Adding Points

Adding Points to a polygonal shape is relatively simple. Immediately after adding the node to the Node Editor, there are no points, but the tool will be in Click Append mode. Click once in the viewer wherever a point is required for the shape. Continue clicking to draw the shape. When the shape is complete, click on the initial point again to close the shape.

When the shape is closed, the mode of the polyline will change to Insert and Modify. This allows for the adjusting and adding of additional points to the shape by clicking on segments of the polyline. To lock down the shape and prevent accidental changes, switch the Polyline mode to Done using the Polyline toolbar or contextual menu.



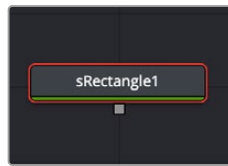
The sPolygon toolbar

When a Polygon (or B-Spline) shape is added to a node, a toolbar appears above the viewer, offering easy access to modes. Hold the pointer over any button in the toolbar to display a tooltip that describes that button's function.

- **Click:** Click is the default option when creating a polyline (or B-Spline) shape. It is a Bézier style drawing tool. Clicking sets a control point and appends the next control point when you click again in a different location.
- **Draw:** Draw is a freehand drawing tool. It creates a shape similar to drawing with a pencil on paper. You can create a new shape using the Draw tool, or you can extend an existing open spline by clicking the Draw tool and starting to draw from the last control point.
- **Insert and Modify:** Insert adds a new control point along the spline and lets you modify it.
- **Modify Only:** Modify allows you to safely move or smooth any exiting point along a spline without worrying about adding new points accidentally.
- **Done:** Prevents any point along the spline from being moved or modified. Also, new points cannot be added. You can, however, move and rotate the entire spline.
- **Close:** Closes an open spline.
- **Smooth:** Changes the selected control point from a linear to a smooth curve.
- **Linear:** Changes the selected control point from a smooth curve to linear.
- **Select All:** Selects all the control points on the spline.
- **Show Key Points:** Shows or hides the control points along the spline.
- **Show All Handles:** Shows or hides the Bézier handles along the polyline.
- **Shape Box:** Places a reshape rectangle around the selected spline shape. Using the reshape rectangle, you can deform groups of control points or entire shapes much easier than modifying each point.
- **Delete Points:** Deletes the selected control point(s).
- **Reduce Points:** Opens a Freehand precision window that can be used to reduce the number of controls points on a spline. This can make the paint stroke easier to modify, especially if it has been created using the Draw tool.

Change the way the toolbar is displayed by right-clicking on the toolbar and selecting from the options displayed in the toolbar's contextual menu.

sRectangle



The sRectangle node

The sRectangle node is used to create rectangular shapes. Like almost all shape nodes, you can only view the sRectangle node's results through a sRender node.

External Inputs

This node generates shapes and does not have any inputs.

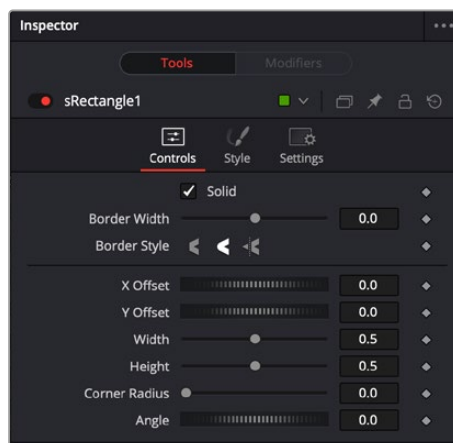
Basic Node Setup

The sRectangle node is a shape generator, meaning it generates a shape and therefore has no inputs. The output of the sRectangle can go into a sRender for viewing and further compositing or, more likely, connect to another Shape node like sGrid or sDuplicate.



An sRectangle node connecting to an sDuplicate node, and then viewed using an sRender node

Inspector



The sRectangle Controls tab

Controls

The Controls tab is used to define the rectangle characteristics, including fill, border, size, and position.

Solid

When enabled, the Solid checkbox fills the rectangle shape with the color defined in the Style tab. When disabled, an outline created by the Border Width control is displayed, and the center is made transparent.

Border Width

This parameter expands or contracts the border around the shape. Although it can be used when the Solid checkbox is enabled, it is primarily used to determine the outline thickness when the checkbox is disabled.

Border Style

The Border Style parameter controls how the sides of the rectangle join at the corners. There are three styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter maintains pointed corners.

Cap style

When the Solid checkbox is disabled, three Cap Style options are displayed. The cap styles can create lines with flat, rounded or squared ends. Flat caps have flat, squared ends, while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The Position parameter is only displayed when the Solid checkbox is disabled. It allows you to position the starting point of the shape. When used in conjunction with the Length parameter, it positions the gap in the outline.

Length

The Length parameter is only displayed when the Solid checkbox is disabled. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the Length parameters allows you to create write-on style animations.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. So an X offset of 0.0 is centered and a value of 0.5 places the center of the shape directly on the right edge of the frame.

Width/Height

The Width and Height parameters determine the vertical and horizontal size of the rectangle. If the values are identical, then you have a square.

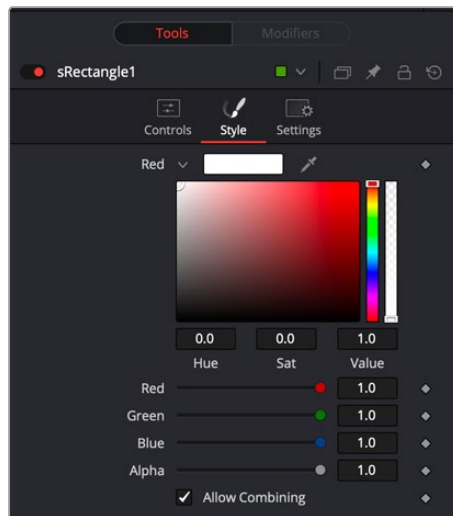
Corner Radius

This parameter determines if the corners of the rectangle are sharp or rounded. A value of 0.0 produces sharp corners, while a value of 1.0 will create a circle from a starting square shape or a pill shape from a rectangle.

Angle

The Angle parameter rotates the shape based on the center axis.

Style Tab



The sRectangle Style tab

Style

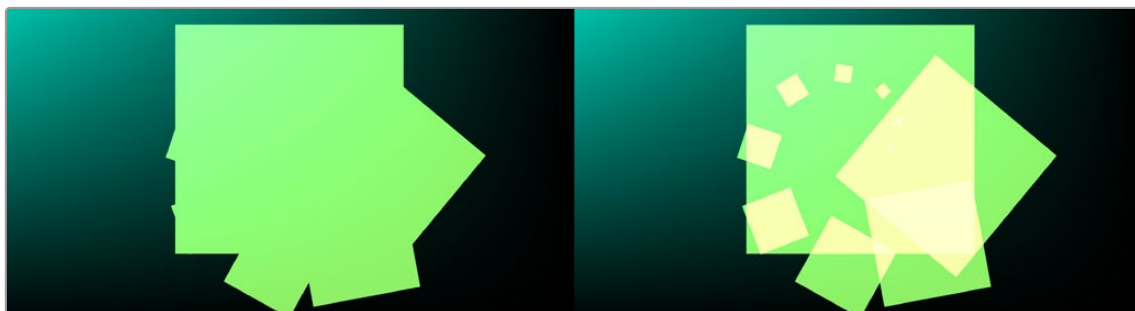
The Style tab is used to assign color to the shape and control its transparency.

Color

The Color parameter controls determine the color of the fill and border from the sRectangle node. To choose a shape color, you can click the color disclosure arrow and use the color swatch, or drag the eye dropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter the value of each color channel or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself. For instance, if a rectangle Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a duplicate or grid node that causes the shape and Alpha channel to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.



Allow Combining Enabled (left), Allow Combining Disabled (right)

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

sRender



The sRender node

The sRender node converts the vector shapes to an image. The output of the sRender allows the vector shapes to be integrated with other elements in a composite.

Inputs

There is one input on the Background node for an Effect Mask input.

Input1: [orange, required] This input accepts the output of your final shape node. A rendered bitmap image is created from the sRender node for compositing into the rest of your comp.

Effect Mask: The optional blue effect mask input accepts a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the displayed area to only those pixels within the mask.

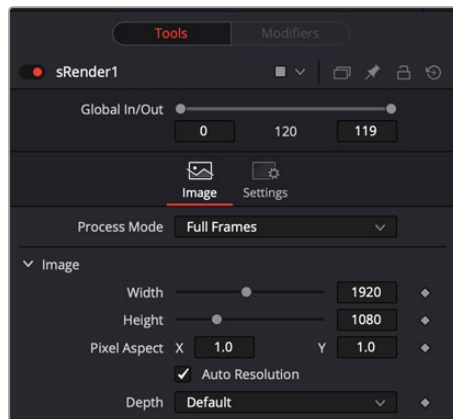
Basic Node Setup

The sRender node is always placed at the end of a string of Shape nodes. The output connects to other 2D nodes like a Soft Glow node.



Multiple Shape nodes connected to the sRender node and then processed and composited with a title

Inspector



The sRender Image tab

Image Tab

The controls in this tab are used to set the resolution, color depth, and pixel aspect of the image produced by the sRender node.

Process Mode

Use this menu control to select the Fields Processing mode used by Fusion to render the resulting image. The default Full Frames option is appropriate for progressive formats.

Width/Height

This pair of controls are used to set the Width and Height dimensions of the image to be created by the sRender node.

Pixel Aspect

This control is used to specify the Pixel Aspect ratio of the created images. An aspect ratio of 1:1 would generate a square pixel with the same dimensions on either side (like a computer display monitor), and an aspect of 0.9:1 would create a slightly rectangular pixel (like an NTSC monitor).

NOTE: Right-click on the Width, Height, or Pixel Aspect controls to display a menu listing the file formats defined in the preferences Frame Format tab. Selecting any of the listed options will set the width, height, and pixel aspect to the values for that format, accordingly.

Auto Resolution

When this checkbox is selected, the width, height, and pixel aspect of the image created by the node will be locked to values defined in the composition's Frame Format preferences. If the Frame Format preferences change, the resolution of the image produced by the node will change to match. Disabling this option can be useful to build a composition at a different resolution than the eventual target resolution for the final render.

Depth

The Depth button array is used to set the pixel color depth of the image created by the Creator node. 32-bit pixels require 4X the memory of 8-bit pixels but have far greater color accuracy. Float pixels allow high dynamic range values outside the normal 0..1 range, for representing colors that are brighter than white or darker than black.

Source Color Space

You can use the Source Color Space menu to set the Color Space of the footage to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion, but rather adds the source space data into the metadata, if that metadata doesn't exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver, if explicit output spaces are defined there. There are two options to choose from:

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Color Space Type menu where you can choose the correct color space of the image.

Source Gamma Space

Using the Curve Type menu, you can set the Gamma Space of the footage and choose to remove it by way of the Remove Curve check box when working in a linear workflow. There are three choices in the Curve Type menu

- **Auto:** Automatically reads and passes on the metadata that may be in the image.
- **Space:** Displays a Gamma Space Type menu where you can choose the correct gamma curve of the image.
- **Log:** Brings up the Log/Lin settings, similar to the Cineon tool.

For more information, see *Chapter 99, "Film Nodes,"* in the DaVinci Resolve Reference Manual, or Chapter 39 in the Fusion Reference Manual.

Remove Curve

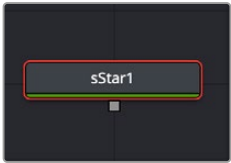
Depending on the selected Gamma Space or on the Gamma Space found in Auto mode, the Gamma Curve is removed from, or a log-lin conversion is performed on, the material, effectively converting it to a linear output space.

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

sStar



The sStar node

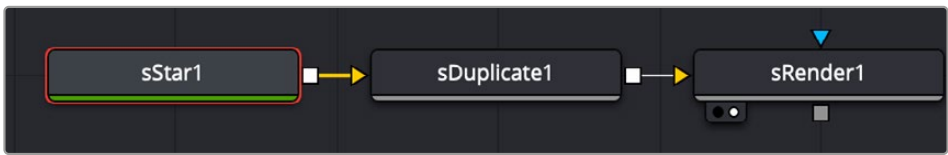
The sStar node is used to create multi-point star shapes. Like almost all Shape nodes, you can only view the sStar node’s results through a sRender node.

External Inputs

This node generates shapes and does not have any inputs.

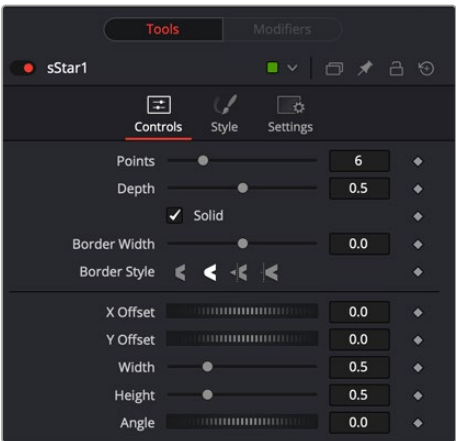
Basic Node Setup

The sStar node is a shape generator, meaning it generates a shape and therefore has no inputs. The output of the sStar can go into a sRender for viewing and further compositing or, more likely, connect to another shape node like sGrid or sDuplicate.



An sStar node connecting to an sDuplicate node, and then viewed using an sRender node

Inspector



The sStar Controls tab

Controls

The Controls tab is used to define the star shape’s characteristics, including number of points, depth, fill, border, size, and position.

Points

This slider determines the number of points or arms on the star.

Depth

The depth slider controls the inner radius or width of the arms. A depth of 0.001 makes hair-thin arms, while a depth of 1.0 makes a faceted circle.

Solid

When enabled, the Solid checkbox fills the star shape with the color defined in the Style tab. When disabled, an outline created by the Border Width control is displayed, and the center is made transparent.

Border Width

This parameter expands or contracts the border around the shape. Although it can be used when the Solid checkbox is enabled, it is primarily used to determine the outline thickness when the checkbox is disabled.

Border Style

The Border Style parameter controls how the sides of the star join at the corners. There are three styles provided as options. Bevel squares off the corners. Round creates rounded corners. Miter maintains pointed corners.

Cap style

When the Solid checkbox is disabled, three cap style options are displayed. The cap styles can create lines with flat, rounded or squared ends. Flat caps have flat, squared ends while rounded caps have semi-circular ends. Squared caps have projecting ends that extend half the line width beyond the end of the line.

The caps are not visible unless the length is below 1.0.

Position

The Position parameter is only displayed when the Solid checkbox is disabled. It allows you to position the starting point of the shape. When used in conjunction with the length parameter, it positions the gap in the outline.

Length

The Length parameter is only displayed when the Solid checkbox is disabled. A length of 1.0 is a closed shape. Setting the length below 1.0 creates an opening or gap in the outline. Keyframing the Length parameters allows you to create write-on style animations.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. So an X offset of 0.0 is centered and a value of 0.5 places the center of the shape directly on the right edge of the frame.

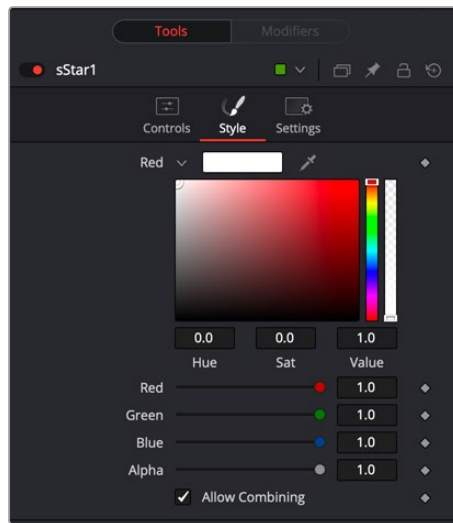
Width/Height

The Width and Height parameters determine the vertical and horizontal size of the star. If the values are identical, then all arms of the star are of equal length.

Angle

The Angle parameter rotates the shape based on the center axis.

Style Tab



The sStar Style tab

Style

The Style tab is used to assign color to the shape and control its transparency.

Color

The Color controls determine the color of the fill and border from the sStar node. To choose a shape color, you can click the color disclosure arrow and use the color swatch, or drag the eye dropper into the viewer to select a color from an image. The RGBA sliders or number fields can be used to enter the value of each color channel or the strength of the Alpha channel.

Allow Combining

When this checkbox is enabled, the Alpha channel value is maintained even when passing through other nodes downstream that may cause the shape to overlap with copies of itself. When disabled, the Alpha channel value may increase when the shape overlaps itself. For instance, if a star Alpha channel is set to .5, enabling the Allow Combining checkbox maintains that value even if the shape passes through a duplicate or grid node that causes the shape and Alpha channel to overlap. Disabling the checkbox causes the Alpha channel values to be compounded at each overlapping area.



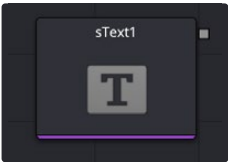
Allow Combining Enabled (left), Allow Combining Disabled (right)

Common Controls

Settings tab

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

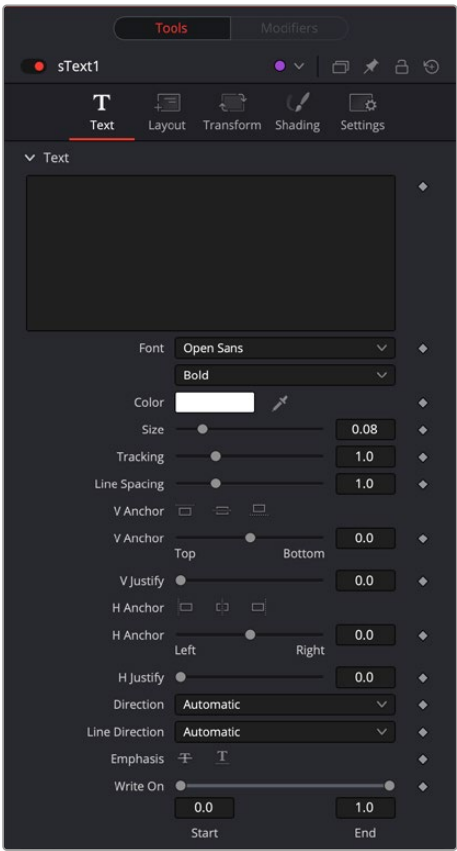
sText [sTxt]



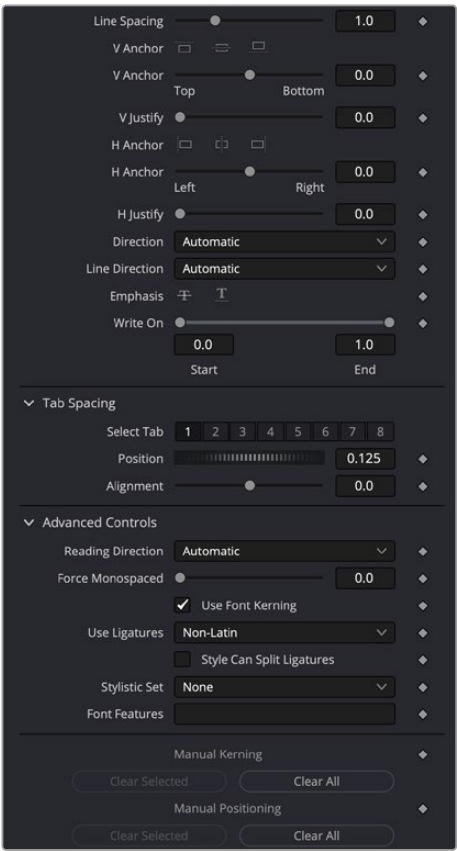
The sText node

Introduction

Designed for the Fusion Shape environment, the sText node is an advanced character generator capable of multiple styles, 3D transformations, and several layers of shading. The sText node is a “Shape” version of the 2D Text+ node. The controls for this node are mostly identical to the controls for the 2D version in almost all respects, except sText doesn’t support gradient colors.



The sText controls



sTransform



The sTransform node

The sTransform node is used to add an additional set of transform controls to the existing controls that are contained in Shape nodes. These additional transforms can be used to create hierarchical animations. For instance, you can use a sStar's built-in Angle control to spin the star around. The star can then be output to an sTransform node. The rotation control in the sTransform can be used to orbit the star around the frame.

Like almost all Shape nodes, you can only view the sStar node's results through a sRender node.

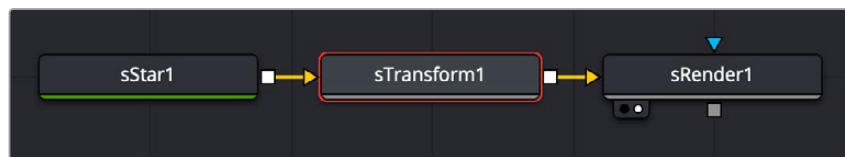
External Inputs

The following input appears on the node's tile in the Node Editor:

Input1: [orange, required] This input accepts the output of another Shape node. The shape connected to this input is moved, scaled, and rotated based on the sTransform settings.

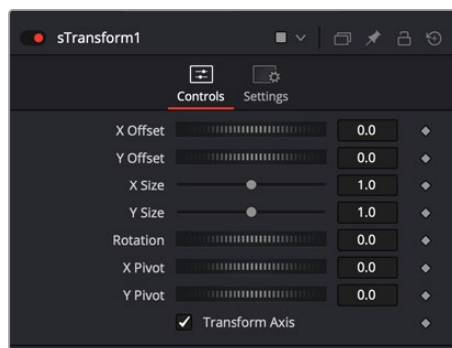
Basic Node Setup

The sTransform node takes the input from another Shape node and adds another set of transforms or hierarchical animation. The output of the sTransform can go into a sRender for viewing and further compositing.



An sStar node connecting to an sTransform node, and then viewed using an sRender node

Inspector



The sTransform Controls tab

Controls

The Controls tab is used to define the add a set of transform controls to the incoming shape.

X and Y Offset

These parameters are used to position the shape left, right, up, and down in the frame. The shape starts in the center of the frame, and the parameters are used to offset the position. The offset coordinates are normalized based on the width of the frame. So an X offset of 0.0 is centered and a value of 0.5 places the center of the shape directly on the right edge of the frame.

X and Y Size

The X and Y Size determine the vertical and horizontal scaling of the incoming shape. If the values are different then the shape will be skewed from its original design.

Rotation

The dial rotates the shape based on the pivot controls.

X and Y Pivot

These parameters position the axis of rotation for the incoming shape. The pivot point is visible in the viewer as a red X. The X can be dragged in the viewer for positioning.

Transform Axis

Check this box to apply the transform to the shape's axis.

Common Controls**Settings tab**

The Settings tab in the Inspector is common to all Shape nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Common Controls

Nodes that handle Shape operations share a number of identical controls in the Inspector. This section describes controls that are common amongst Shape nodes.

Settings Tab

The Settings tab in the Inspector can be found on every Shape node. Most of the controls listed here are only found in the sRender node but a few are common to all Shape nodes.

Blend (sRender only)

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming. Normally, this causes the tool to skip processing entirely, copying the input straight to the output.

Process When Blend Is 0.0 (sRender only)

The tool is processed even when the input value is zero. This can be useful if processing of this node is scripted to trigger another task but the value of the node is set to 0.0.

Red/Green/Blue/Alpha Channel Selector (sRender only)

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the red button on a Blur tool is deselected, the blur is first applied to the image, then the red channel from the original input is copied back over the red channel of the result.

There are some exceptions, such as tools where deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Control tabs are identical.

Apply Mask Inverted (sRender only)

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

Multiply By Mask (sRender only)

Selecting this option causes the RGB values of the masked image to be multiplied by the mask channel's values. This causes all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

Motion Blur (sRender only)

- **Motion Blur:** This toggles the rendering of motion blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 causes Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one whole frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

Use GPU (sRender only)

The user GPU menu has three settings. Setting the menu to Disable turns off hard accelerated rendering using the graphics card in your computer. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the empty field will hide that incoming connection line as long as the node is not selected in the node tree.. When the node is selected in the node tree the line will reappear.

Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower left corner of the node when the full tile is displayed or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more information on scripting nodes, see the Fusion scripting documentation.